University Institute of Engineering

Department of Computer Science & Engineering

COMPUTER ORGANIZATION & ARCHITECTURE

(23CST-204/23ITT-204)

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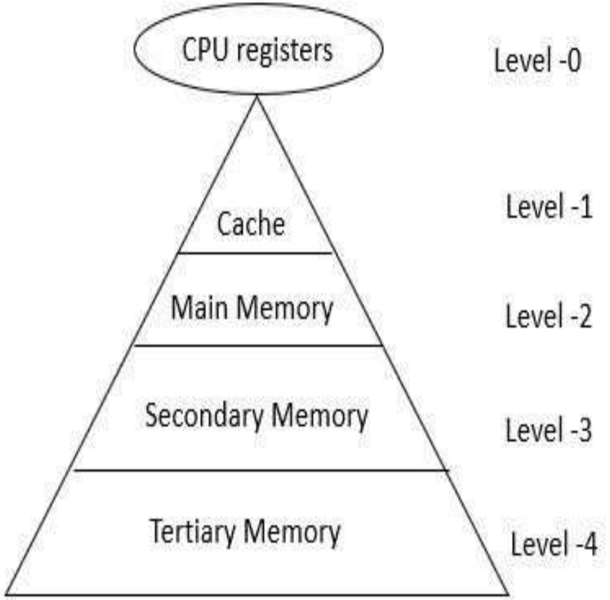
MEMORY HIERARCHY

The computer memory hierarchy looks like a pyramid structure which is used to describe the differences among memory types. It separates the computer storage based on hierarchy.

Level 0: CPU registers
Level 1: Cache memory
Level 2: Main memory or primary memory
Level 3: Magnetic disks or secondary memory
Level 4: Optical disks or magnetic types or tertiary Memory



In Memory Hierarchy the cost of memory, capacity is inversely proportional to speed. Here the devices are arranged in a manner Fast to slow, that is from register to Tertiary memory.

Let us discuss each level in detail:

Level-0 (Registers)

The registers are present inside the CPU. As they are present inside the CPU, they have least access time. Registers are most expensive and smallest in size generally in kilobytes. They are implemented by using Flip-Flops.

Level-1 (Cache Memory)

Cache memory is used to store the segments of a program that are frequently accessed by the processor. It is expensive and smaller in size generally in Megabytes and is implemented by using static RAM.

Level-2 (Primary or Main Memory)

It directly communicates with the CPU and with auxiliary memory devices through an I/O processor. Main memory is less expensive than cache memory and larger in size generally in Gigabytes. This memory is implemented by using dynamic RAM.

Level-3 (Secondary storage)

Secondary storage devices like Magnetic Disk are present at level 3. They are used as backup storage. They are cheaper than main memory and larger in size generally in a few TB.

Level-4 (Tertiary storage)

Tertiary storage devices like magnetic tape are present at level 4. They are used to store removable files and are the cheapest and largest in size (1-20 TB).

Let us see the memory levels in terms of size, access time, bandwidth.

Level	**Register**	**Cache memory**	**Primary memory**	**Secondary memory**
Bandwidth	4k to 32k MB/sec	800 to 5k MB/sec	400 to 2k MB/sec	4 to 32 MB/sec
Size	Less than 1KB	Less than 4MB	Less than 2 GB	Greater than 2 GB
Access time	2 to 5 nsec	3 to 10 nsec	80 to 400 nsec	5 ms
Managed by	Compiler	Hardware	Operating system	OS or user

Why is Memory Hierarchy used in systems?

Memory hierarchy is arranging different kinds of storage present on a computing device based on speed of access. At the very top, the highest performing storage is CPU registers which are the fastest to read and write to. Next is cache memory followed by conventional DRAM memory, followed by disk storage with different levels of performance including SSD, optical and magnetic disk drives.

To bridge the processor memory performance gap, hardware designers are increasingly relying on memory at the top of the memory hierarchy to close/reduce the performance gap. This is done through increasingly larger cache hierarchies (which can be accessed by processors much faster), reducing the dependency on main memory which is slower.

This Memory Hierarchy Design is divided into 2 main types:

- **External Memory or Secondary Memory –**
  Comprising of Magnetic Disk, Optical Disk, Magnetic Tape i.e. peripheral storage devices which are accessible by the processor via I/O Module.

- **Internal Memory or Primary Memory –**
  Comprising of Main Memory, Cache Memory & CPU registers. This is directly accessible by the processor.

These are the following characteristics of Memory Hierarchy Design from above figure:

1. Capacity:

It is the global volume of information the memory can store. As we move from top to bottom in the Hierarchy, the capacity increases.

2. Access Time:

It is the time interval between the read/write request and the availability of the data. As we move from top to bottom in the Hierarchy, the access time increases.

3. Performance:

Earlier when the computer system was designed without Memory Hierarchy design, the speed gap increases between the CPU registers and Main Memory due to large difference in access time. This results in lower performance of the system and thus, enhancement was required. This enhancement was made in the form of Memory Hierarchy Design because of which the performance of the system increases. One of the most significant ways to increase system performance is minimizing how far down the memory hierarchy one has to go to manipulate data.

4. Cost per bit:

As we move from bottom to top in the Hierarchy, the cost per bit increases i.e. Internal Memory is costlier than External Memory.

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- J.P. Hayes, “Computer Architecture and Organization”, Third Edition.
- Mano, M., “Computer System Architecture”, Third Edition, Prentice Hall.
- Stallings, W., “Computer Organization and Architecture”, Eighth Edition, Pearson Education.

Text Books:

- Carpinelli J.D,” Computer systems organization &Architecture”, Fourth Edition, Addison Wesley.
- Patterson and Hennessy, “Computer Architecture”, Fifth Edition Morgaon Kauffman.

Reference Website

- Memory Hierarchy Design and its Characteristics - GeeksforGeeks
- What is memory hierarchy? (tutorialspoint.com)